

Making a Move

Once our brain has planned a movement (see pp.96–97), it sends signals to the appropriate muscles in the body, via the nervous system, to turn intention into action.

From brain to spine

Signals from the motor and parietal areas of the cortex are sent along the axons of neurons, through the brain stem, to communicate with motor neurons in the spinal cord. Most of the axons form part of a bundle called the lateral corticospinal tract, which crosses over at the base of the brain stem so that axons from one brain hemisphere connect to motor nerves for the opposite side of the body. Other nerve tracts originate in different parts of the midbrain and perform specific movement functions.



1 Nerve tracts

The axons of the lateral corticospinal tract send signals to muscles that connect to the skeleton to produce voluntary limb movements. Other groups of axons are responsible for the body's involuntary responses, such as balance, as well as for fine-tuning movements.

SIMPLE AND COMPLEX MOVEMENTS

A motor homunculus shows which areas of the motor cortex control which areas of the body. Areas for adjacent body parts—such as the arm and hand—are generally grouped together. The body parts are shown in proportion; those areas that make complex movements, such as the face and the hand, take up more space in the cortex than those making simple movements, such as the foot.

